



EMBEDDED SYSTEMS

ARM Cortex-M // RTOS // Automotive // Edge AI

22

WEEKS

40+

SESSIONS

06

CAPSTONES

100%

PLACEMENT AID

STM32

FreeRTOS

Embedded C

CAN / LIN

AUTOSAR

TinyML

Embedded Linux

Zephyr

IoT / BLE

Secure Boot

OUR ALUMNI WORK AT

Microsoft • BOSCH • IBM • EY • Capgemini • Deloitte • Adobe

WHO ARE WE?



Edufyi Tech Solutions builds job-ready engineers. Our virtual internships and industry-mentored training programs pair deep technical curriculum with six real-time and capstone projects per domain, so every learner graduates with a portfolio, not just a certificate.

WHY CHOOSE US



SELF-PACED

Choose your speed, master the skills with flexible cohorts.



PROJECT LAB

Industry project playground on real hardware & simulators.



MENTORS

Live mentorship from MNC engineers and startup founders.



PLACEMENT

100% placement assistance, unlimited mocks and GDs.



CERTIFIED

Wipro DICE ID accredited, globally recognised certification.



LIFETIME LMS

Personalised LMS with recorded lectures, anytime access.



R&D CONTENT

Exclusive research-grade content from leading MNCs.



STIPEND

Performance-based stipend plus 1:1 career counselling.

ALSO INCLUDED

Unlimited mocks & GDs

24x7 dedicated support

Aptitude grooming

Resume building

Recorded lectures

Hiring partner access

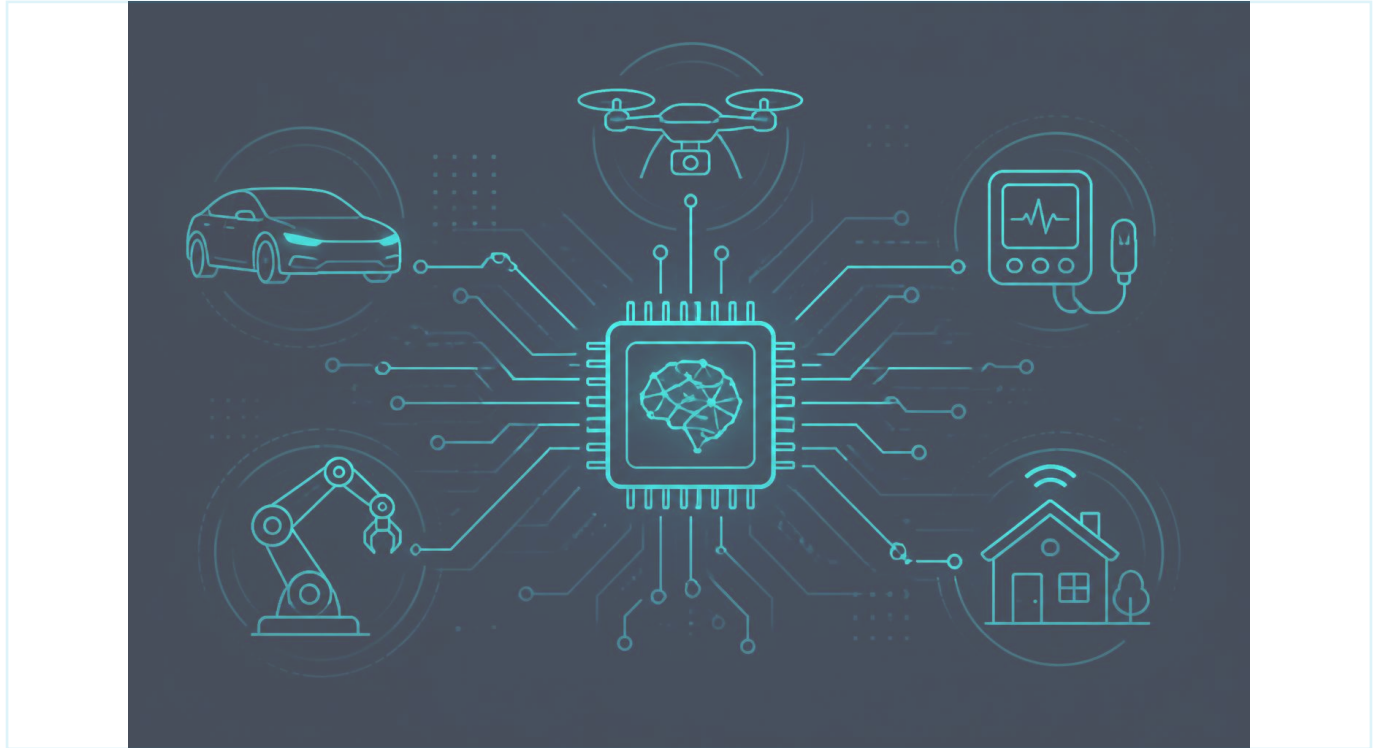
Startup founder AMAs

Domain electives

"We don't teach slides. We teach the firmware that ships."

WHAT IS AN EMBEDDED SYSTEM?

A purpose-built fusion of hardware and firmware that senses, decides and actuates in real time – inside cars, pacemakers, satellites, factory robots, wearables and every smart device shipping today.



SCOPE & MARKET

\$116.2B

market size by 2030

16.1%

CAGR through 2030

30B+

connected edge devices

#1

hiring core-tech domain

WHERE THE DEMAND IS

Automotive & EV

Aerospace & Defence

Medical Devices

Industrial IoT 4.0

Consumer Electronics

Robotics & Drones

Semiconductor & VLSI

Smart Energy

ADVANCED PROGRAM CURRICULUM

MODULE 01

FOUNDATIONS & EMBEDDED C

- Embedded system anatomy, classification, real-time vs general purpose computing
- Microprocessor vs microcontroller architecture, Harvard vs Von Neumann, pipelines
- Advanced Embedded C: pointers, bit manipulation, volatile/const, memory qualifiers
- Data structures for constrained targets: ring buffers, state machines, lookup tables
- Toolchain deep dive: GCC/ARM, linker scripts, startup code, map files, .bss/.data
- Version control, Makefiles/CMake and modular firmware project structure

MODULE 02

ARM CORTEX-M & BARE-METAL DRIVERS

- ARM Cortex-M0+/M4/M7 core, registers, Thumb-2, CMSIS and memory-mapped I/O
- Register-level GPIO, clock tree & PLL configuration, power modes and low-power design
- NVIC, interrupts, exception handling, priorities and vector table relocation
- Timers, PWM, input capture, watchdog, RTC, DMA controller and double buffering
- ADC/DAC, analog front-end, sensor signal conditioning and calibration
- Writing a reusable HAL from the datasheet + reference manual

MODULE 03

COMMUNICATION PROTOCOLS

- UART framing, baud calculation, parity, DMA-driven serial and error handling
- I2C: arbitration, clock stretching, multi-master buses, pull-up sizing
- SPI & QSPI: modes, external flash, display and sensor interfacing
- CAN & CAN-FD: arbitration, DBC signals, bus-off recovery, automotive gateways
- LIN, RS-485/Modbus RTU and industrial fieldbus basics
- USB CDC/HID, Ethernet, MQTT, BLE 5.x, LoRaWAN and Wi-Fi provisioning
- Protocol debugging with logic analyser, oscilloscope and bus decoders

ADVANCED PROGRAM CURRICULUM

MODULE 04

RTOS & REAL-TIME ENGINEERING

- RTOS internals: scheduler, context switch, tick, priority-based preemption
- FreeRTOS & Zephyr: tasks, queues, semaphores, mutexes, event groups, timers
- Priority inversion, deadlocks, race conditions and inheritance protocols
- Rate monotonic analysis, WCET, jitter, latency budgeting and profiling
- Inter-processor communication, tickless idle and RTOS power management
- Stack usage analysis, heap policies and hard-fault post-mortem debugging

MODULE 05

EMBEDDED LINUX & SoC PLATFORMS

- Boot chain: ROM -> U-Boot -> kernel -> root filesystem, device tree overlays
- Yocto & Buildroot image builds, cross compilation and BSP customisation
- Linux character device drivers, sysfs, GPIO/I2C/SPI subsystems
- Raspberry Pi / BeagleBone / i.MX platform bring-up and peripheral control
- Multithreading, IPC, systemd services and containerised edge workloads
- PREEMPT_RT real-time patch and performance tuning

MODULE 06

AUTOMOTIVE & FUNCTIONAL SAFETY

- Automotive E/E architecture, ECUs, zonal & domain controllers
- AUTOSAR Classic layered architecture, BSW, RTE and SWC configuration
- UDS diagnostics (ISO 14229), OBD-II, DTCs and flashing over CAN
- ISO 26262 functional safety lifecycle, ASIL levels, FMEA and safety mechanisms
- MISRA C compliance, static analysis and code coverage for safety-critical firmware
- EV powertrain, BMS fundamentals, motor control and sensor fusion basics

ADVANCED PROGRAM CURRICULUM

MODULE 07

EDGE AI, IoT & TinyML

- TinyML pipeline: data capture, feature engineering, training, quantisation
- TensorFlow Lite Micro & CMSIS-NN deployment on Cortex-M targets
- Anomaly detection, keyword spotting, gesture and vibration classification
- Sensor fusion with IMUs, Kalman/complementary filters
- Edge-to-cloud: MQTT/TLS, AWS IoT Core, digital twins, telemetry dashboards
- OTA firmware update design, delta updates and rollback strategy

MODULE 08

SECURITY, TESTING & PRODUCTION

- Embedded security threat modelling, secure boot chain and chain of trust
- Cryptography on MCUs, TRNG, key storage, TrustZone-M and secure elements
- Debug port locking, readout protection and firmware encryption
- Unit testing with Unity/CMock, HIL testing and CI for firmware
- Hardware bring-up, schematic reading, PCB basics, EMI/EMC and DFM
- Documentation, release engineering and production programming

MODULE 09 / ELECTIVE TRACKS

CHOOSE YOUR SPECIALISATION

Pick one elective track in the final phase and build your sixth capstone inside it, mentored by an engineer working in that industry today.

Automotive & AUTOSAR

Robotics & Drone Control

Edge AI / TinyML

Embedded Linux & BSP

Medical Device Firmware

VLSI & FPGA Basics

Industrial IoT 4.0

ASSESSMENT MODEL

Every module closes with a graded quiz, a hands-on lab review and a mentor code walkthrough. Firmware is reviewed the way it is in industry: readability, memory footprint, timing behaviour and MISRA compliance.

TOOLS YOU WILL MASTER

Every module is taught on the same tooling used by product teams in automotive, medical and industrial firmware organisations.

SILICON & BOARDS

STM32F4 / STM32H7

ESP32-S3

Arduino Uno / Nano

Raspberry Pi 4

TI MSP430

nRF52 BLE

Renesas RA

IDEs & TOOLCHAINS

STM32CubeIDE

Keil uVision

VS Code + PlatformIO

ARM GCC

Yocto / Buildroot

Git & CI

DEBUG & TEST

ST-Link / J-Link

GDB & OpenOCD

Logic Analyser

Oscilloscope

CANoe / BusMaster

Unity + CMock

SIMULATION & AI

Proteus

Tinkercad

MATLAB / Simulink

QEMU

TensorFlow Lite Micro

Edge Impulse

SIX INDUSTRY CAPSTONES

P1

CAN-BUS VEHICLE NODE

Multi-ECU CAN network with UDS diagnostics and a live dashboard.

P2

RTOS SMART CONTROLLER

FreeRTOS multi-task controller with sensor fusion and OTA updates.

P3

TinyML PREDICTIVE MAINT.

Vibration anomaly detection running on a Cortex-M4 in real time.

P4

IoT ENERGY MONITOR

BLE + MQTT telemetry to cloud with TLS and digital-twin dashboard.

P5

EMBEDDED LINUX GATEWAY

Custom Yocto image, device driver and edge-to-cloud bridge.

P6

SECURE BOOTLOADER

Signed firmware, encrypted delta OTA and safe rollback on STM32.

CAREER OPPORTUNITIES

Embedded Software Engineer

4.5 - 12 LPA

Firmware for consumer, industrial and medical products

Firmware Developer (RTOS)

5 - 14 LPA

Real-time task design on FreeRTOS / Zephyr targets

Automotive / AUTOSAR Engineer

6 - 16 LPA

ECU software, CAN networks, UDS diagnostics, ISO 26262

IoT & Edge Solutions Engineer

5 - 13 LPA

Connected devices, BLE / MQTT, cloud telemetry

Embedded Linux / BSP Engineer

7 - 18 LPA

Yocto builds, device drivers, SoC bring-up

Hardware Validation & Test Engineer

4 - 10 LPA

HIL rigs, bench automation, protocol compliance

Robotics / Control Systems Engineer

6 - 15 LPA

Motion control, sensor fusion, ROS integration

Embedded Security Engineer

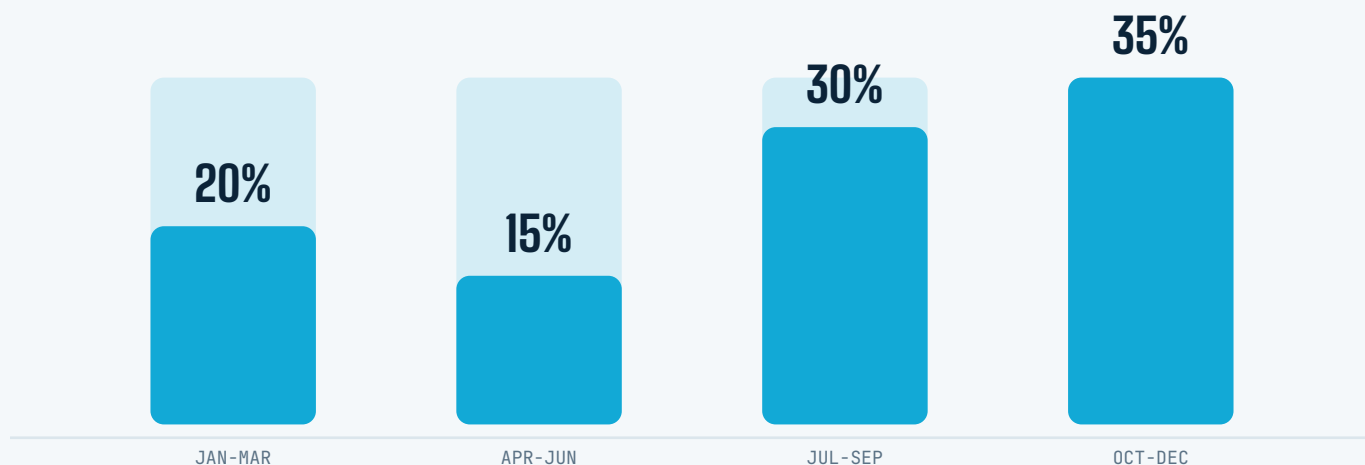
8 - 20 LPA

Secure boot, TrustZone-M, crypto and key management

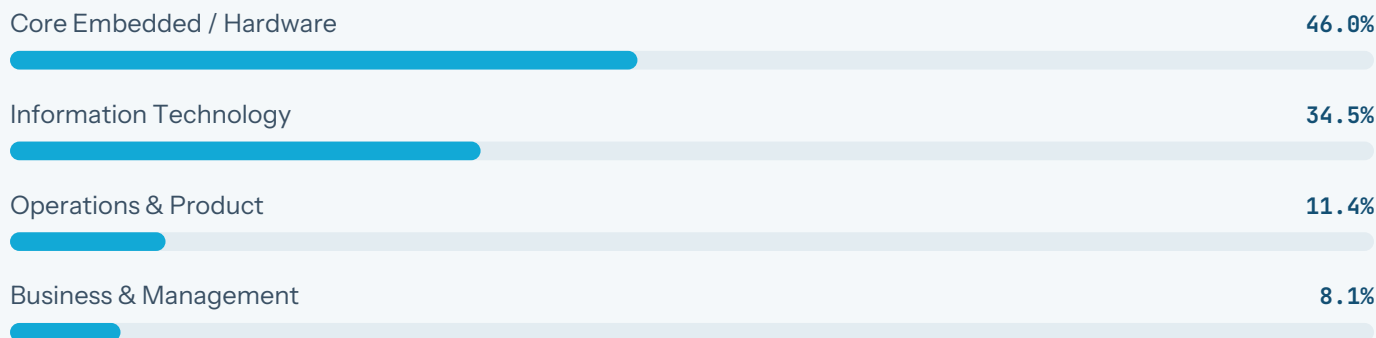
Indicative entry-level ranges for the Indian market; actual offers vary by company, location and interview performance.

PLACEMENT ANALYTICS

QUARTERLY PLACEMENT ANALYSIS



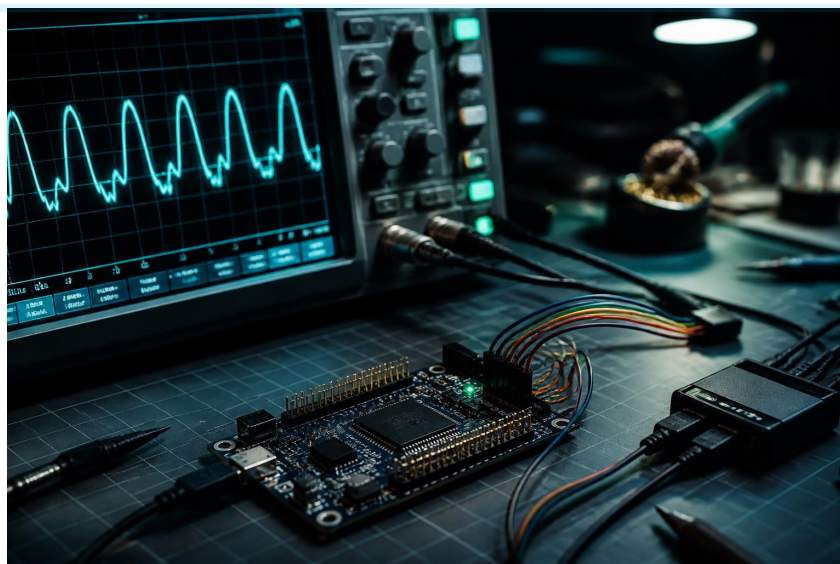
OFFERED PROFILES VIA EDUFYI



LEARNER'S BACKGROUND



HOW YOUR JOURNEY WORKS



01

ENROL

Pick your track and pace with a 1:1 counselling session.

02

LEARN

Live mentor-led sessions plus lifetime LMS access.

03

BUILD

Six advanced capstones on real hardware and simulators.

04

CERTIFY

Wipro DICE ID accredited plus MNC certifications.

05

GET HIRED

Resume, mocks, GDs and hiring-partner referrals.

CERTIFICATIONS YOU EARN

WIPRO DICE ID

Globally recognised, blockchain-verified accreditation

INTERNSHIP CERTIFICATE

Domain internship completion with project record

INDUSTRIAL TRAINING

Signed industrial training certificate

MNC CERTIFICATIONS

Microsoft, Cisco, IBM, Autodesk, Adobe, ESB

AUTHORISED CERTIFICATION PARTNERS

Adobe • Microsoft • QuickBooks • Autodesk • Cisco • ESB

ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



INTERNSHIP CERTIFICATE
Accredited Certification



INDUSTRIAL TRAINING
Accredited Certification



PROJECT COMPLETION
Accredited Certification

INTERNSHIP

Verified internship completion credential

TRAINING

Industrial training hours certified

PROJECT

Capstone project completion proof

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



AUTHORISED PARTNERS.

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



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MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

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